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Numerical prediction of unsteady vortex shedding for large leading-edge roughness

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Abstract

A full two-dimensional Navier–Stokes algorithm is used to investigate unsteady, incompressible viscous flow past an airfoil leading edge with surface roughness that is characteristic of ice accretion. The roughness is added to the surface through the use of a Prandtl transposition and can generate both small-scale and large-scale roughness. The focus of the study is a detailed flow analysis of the unsteady velocity fluctuations and vortex shedding induced by the surface roughness. The results of this study are compared to experimental data on roughness-induced transition for the same roughness geometry. A comparison is made between "fluctuation intensity" values from the current algorithm to experimentally determined turbulence intensity values. The effects of the roughness Reynolds number, Re_k, are investigated and compared to experimental values of the critical roughness Reynolds number. The authors speculate that there may be a possible correlation between unsteady roughness-induced vortex shedding and the onset of experimentally measured transitional flow downstream of large-scale roughness.
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1. Introduction

Over the last 60 years, there has been a concentrated effort to investigate how surface roughness impacts the characteristics of fluid flow. Surface roughness in the aerospace community is a broad term encompassing many different sizes and shapes of roughness, including the almost non-detectable roughness such as machining-process imperfections on an airfoil up to large-scale

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roughness such as ice accretion geometries. In recent years, aircraft icing, which can produce a severe type of surface roughness, has emerged as an important field of study with an increased number of aviation accidents being attributed to adverse icing conditions. One particular area of interest in both aircraft icing and the general surface roughness community is the role that roughness plays in promoting premature transition to turbulence (e.g. [1–4]). This type of transition, termed roughness-induced transition, is a complex flow phenomenon, and though research has been conducted to help understand the mechanisms involved in this process, there are still many gaps in the knowledge base for a complete understanding.

The present work investigates unsteady viscous flow about an airfoil leading edge containing large-scale surface roughness with a focus on the resulting velocity fluctuations and vortex shedding. The airfoil roughness is characteristic of early growth glaze ice geometries. It has been documented [5–7] that leading-edge surface roughness, in particular ice roughness, can adversely effect the aerodynamic characteristics and performance of a wing (or airfoil). This can include loss of lift, increase in drag, and alterations to the pitching moment. The severity of the degradation is dependent upon many factors, including the extent, size, and location of the roughness. The results of this study are compared to experimental results of roughness-induced transition for the same type of surface roughness. The authors speculate that there may be a possible correlation between unsteady roughness-induced vortex shedding and the onset of experimentally measured transitional flow downstream of the same roughness. Therefore, a brief review of research efforts involving surface roughness and transitional flow is warranted.

Investigations have been conducted into the mechanisms by which roughness enhances transitional flow (e.g. [1–4]), and it is clear that some surface roughness can promote transition in a different manner than "natural" transition occurs for flow past a smooth surface. For many types of surface roughness, in particular, roughness that protrudes through the boundary layer, the exact transition mechanism is unknown. This type of transition bypasses known linear transition mechanisms and is therefore referred to as bypass transition [8]. Kerho [9] showed that the process of large-scale roughness-induced transition was significantly different than that of the T–S transition mechanism for a smooth surface.

One area where the effects of small-scale roughness on transition have been investigated is in the receptivity mechanism. Morkovin [10] has defined receptivity as the process in which external disturbances enter the boundary layer. Goldstein [11] conducted work that showed small-scale surface roughness can couple with an externally imposed acoustic disturbance to form a combined receptivity mechanism. The roughness acts to transfer the relatively long wavelength of the sound waves into shorter wavelengths of T–S waves that initiate transition. Wu [12] conducted a similar investigation and showed that boundary layer receptivity can also develop from interaction of small-scale roughness and vortical freestream disturbances. This interaction of disturbances with small-scale variations in surface geometry is sometimes termed "localized receptivity" [13]. Examining the available literature, it is clear that some types of surface roughness (small-scale) can take part in the receptivity process and impact the linear instability mechanisms such as T–S waves, as described in the references above. However, there are other types of roughness geometries (large-scale) that produce a "bypass transition" [8] where the typical mechanism of T–S waves initiating the transition process is bypassed.

Previously published data (e.g. [14,15]) have shown that separation due to roughness can have a destabilizing effect and amplify the instability of the boundary layer. Fasel et al. [16] made similar

conclusions, but also speculated that the roughness element appeared to act as a source of additional disturbances which were produced by the vortex shedding mechanism. More recently, Bragg et al. [17] conducted wind tunnel experiments for large-scale roughness on the leading edge of an airfoil. The authors commented that the roughness-induced transition process appeared to be due to vortex shedding and breakdown.

The degree to which roughness impacts the transition mechanism is dependent in part on the type of roughness and the relative height of the roughness with respect to the boundary layer. A majority of the available experimental data for roughness-induced transition focus on roughness element heights that are smaller than the approaching laminar boundary layer thickness and with a zero pressure gradient (flat plate). But, for a typical leading-edge ice accretion, the roughness quickly grows large enough to protrude through the boundary layer. The roughness geometry employed in this work matches the roughness length, height and placement used by Bragg et al. [20], which does protrude through the boundary layer (i.e. large leading-edge roughness). The roughness heights and locations, for Bragg et al.'s [20] study as well as the current study, are based on experimental ice accretion work done by Shin [21] for early ice growth on a NACA 0012 airfoil.

A thorough review of the current understanding for the effects of roughness on boundary-layer transition is presented by Bragg et al. [18] and Winkler [19]. Kerho and Bragg [4] make the observation that no direct modeling or predictive capability is currently available for this type of complex, non-linear process. Therefore, with the exception of a full three-dimensional DNS study, there are few resources available that provide a reliable method for predicting roughness-induced transition.

Obviously, it would be useful to have the capability to predict the location where the flow is laminar, transitional, and fully turbulent, particularly when surface roughness is present. Otherwise, one can only speculate on flow regime locations without detailed experimental validation work. For example, it was thought by Hansman et al. [22] and Yamaguchi and Hansman [23] that a complete transition to turbulent flow occurred at the smooth-rough geometrical boundary that develops in some ice accretion scenarios. However, recent experimental work by Kerho [9] and Bragg et al. [20] have shown that this is typically not the case. Very often there exists a laminar region near the stagnation point even when ice roughness is present on the leading edge. It was found that the roughness elements did trigger the onset of the transition process (at or very near the trailing edge of the roughness), but the transitional flow extended far downstream prior to the flow becoming fully turbulent. In this case, it took a detailed experiment to show the regions of laminar, transitional, and fully turbulent flow.

The work reported in this paper involves the investigation of unsteady flows with surface ice geometries present near the leading edge of an airfoil. The algorithm was developed to study the effects of surface roughness on leading-edge flow and is capable of capturing various roughness-induced flow phenomena. The flow field is modeled using the full unsteady, two-dimensional, incompressible Navier–Stokes equations. The results from this work are compared to experimental data for unsteady transitional flow past surface ice geometries [18,20].

Though fully turbulent flow is three dimensional in nature, and much of the transitional flow regime exhibits some three-dimensionality, it may be possible to predict the onset and early stages of transitional flow with a two-dimensional analysis. Some work has previously been done in numerical prediction of transitional flow with a two-dimensional Navier–Stokes algorithm.

One such study, reported by Fasel [24], showed that the two-dimensional Navier–Stokes algorithm used in their study was capable of predicting stability and initial transition for incompressible, two-dimensional boundary-layer flows. In addition, Fasel et al. [16] investigated the effect of a single roughness element on transition using a two-dimensional Navier–Stokes algorithm.

It is expected that there would be differences between a two-dimensional and three-dimensional analysis of flow separation and vortex shedding. The vortical structures in a three-dimensional flow may be different in size and strength as a result of the ability of the fluid to dissipate some of the energy over the third dimension. Experimental work on unsteady flow separation past an ice shape was conducted by Gurbacki and Bragg [25]. A comparison was made between a two-dimensional and three-dimension ice shape, which revealed a shorter length separation bubble downstream of the three-dimensional shape. It should be noted that these two ice shapes had a peak roughness height that was significantly larger than the roughness height used in the present study.

The surface roughness for the current work is incorporated onto the leading edge using a Prandtl transposition, which allows for rapid analysis of different roughness geometries. This method of roughness generation can produce isolated humps, distributed roughness, or more complex roughness shapes as shown in the work by Huebsch and Rothmayer [26]. Two types of surface roughness are investigated in this work, including an isolated roughness element and distributed roughness, both of which are two-dimensional. One objective of this study is to compare results to experimental work, so the roughness geometries were chosen to match the three-dimensional hemispheres tested by Bragg et al. [20], but in a two-dimensional sense. This limitation may effect the comparison depending on the relative size of the roughness. Kerho [9] did note some similarities for roughness-induced transition between a two-dimensional boundary-layer trip and three-dimensional distributed roughness of the same relative height (and similar to the roughness used in this study), which may indicate a corresponding mechanism for roughness of this height. Morkovin [2] postulated that the leading and trailing edges of three-dimensional distributed roughness can form quasi-two-dimensional protuberances, and therefore, the two-dimensional roughness transition mechanism could be active to a certain extent.

In calculating the unsteady flow with vortex shedding, the solver uses a local-flow analysis from the standpoint that the simulations focus on the airfoil leading-edge region. A semi-infinite parabola is used to simulate the leading-edge region of the airfoil. Restricting the flow analysis to the leading-edge region of an airfoil (or wing) is prompted by the following reasons: (1) the most significant ice accretion and surface roughness occur in this location, (2) this region is susceptible to many important flow phenomena, including leading-edge stall, (3) the region close to the leading edge tends to remain laminar and at a low local Reynolds number, which simplifies the direct simulations, and (4) focusing on the leading-edge region significantly reduces the computational expense. This produces a solver that is clean and robust with no requirement for explicit artificial dissipation or other dissipation-inducing numerical devices. Solving the unsteady flow field only around the leading-edge, and not including the entire airfoil, also allows the use of a large number of grid points in the area of interest. In the current study, a dense grid is placed on the leading edge, which is capable of capturing small eddies within the boundary layer (i.e. in a manner similar to a full DNS code except for the two-dimensional constraint).

2. Governing equations

The flow past the parabola is modeled by the full Navier–Stokes equations for two-dimensional, unsteady, incompressible flow. The x- and y-coordinates are non-dimensionalized by the leading-edge radius of curvature of the parabola, l, the u and v velocities by the freestream velocity, V_∞, and the time, t, by l/V_∞. The resulting dimensionless Navier–Stokes equations cast in the Cartesian coordinate system and written in streamfunction–vorticity form are given as follows:

$$\psi_{xx} + \psi_{yy} = -\omega, \tag{1}$$

$$\omega_t + \psi_y \omega_x - \psi_x \omega_y = Re^{-1}(\omega_{xx} + \omega_{yy}), \tag{2}$$

where ψ and ω are the two-dimensional streamfunction and vorticity, respectively.

The leading edge of the airfoil is approximated by using a semi-infinite parabola, in a manner similar to work by Werle and Davis [27] and Davis [28]. The results for flow past a parabola compare reasonably well with computational results for the NACA 0012 airfoil [29,30] and may be used to simulate any of the NACA 00XX series with different thicknesses since the leading edges of these airfoils are nearly parabolic in shape. A comparison of the leading edge of the parabola and the NACA 0012 airfoil is shown in Fig. 1. The parabola matches the airfoil up to approximately 6% of chord. The reference coordinate system is attached to the parabola and therefore is non-inertial. Since the parabola does not undergo any pitchup maneuvers in this study, the non-inertial terms are ignored.

It is advantageous to cast the governing equations in a parabolic coordinate system in order to produce an orthogonal body-fitted grid. The parabolic coordinate system is related to the Cartesian system through the following transformations:

$$x = \frac{\xi'^2 - \eta'^2}{2}, \quad y = \xi'\eta'. \tag{3}$$

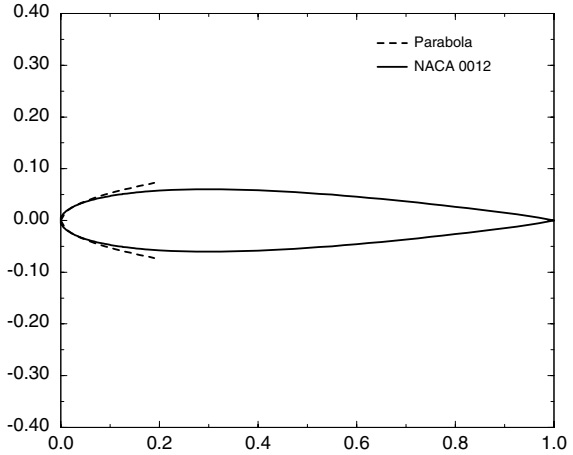


Fig. 1. Comparison of the leading-edge region for the NACA 0012 and the parabola.

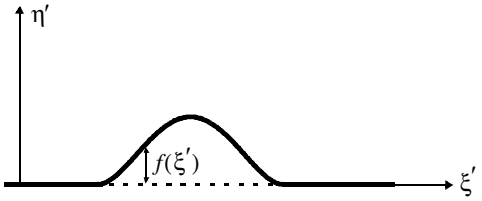


Fig. 2. Schematic diagram of the shearing transformation (or Prandtl transposition).

The surface roughness on the leading edge of the parabola is introduced as a perturbation to the baseline parabola using a shearing transformation (or Prandtl transposition) similar to work reported by Lynn and Rothmayer [31]. The shearing transformation allows the surface perturbation to be directly embedded within the governing equations. This type of analysis can be used on a wide range of single-valued roughness geometries with no need for re-gridding, even when the surface geometry changes. The shearing transformation is depicted in Fig. 2 and given by:

$$\xi' = \xi, \quad \eta' = \eta + f(\xi), \tag{4}$$

where $f(\xi)$ is a known analytic expression. The unsteady flow with vortex shedding investigated in the present study produces high vorticity gradients. To resolve these steep gradients within localized regions near the leading edge, grid stretching and clustering is also applied to the governing equations and the transformation is given by $\bar{\xi} = \bar{\xi}(\xi)$ and $\bar{\eta} = \bar{\eta}(\eta)$.

The transformation from the uniform-grid computational space out to the physical space of the parabola involves multiple steps that are outlined above. The overall transformation is summarized in Fig. 3. A more detailed description of this derivation is given by Huebsch and Rothmayer [26].

In the work reported by Bhaskaran and Rothmayer [32,33], it was found that the convergence rate for a streamfunction–vorticity formulation was improved if the flow variables were split into an inviscid part and a viscous part (i.e. as you move away from the wall, the viscous effects die out

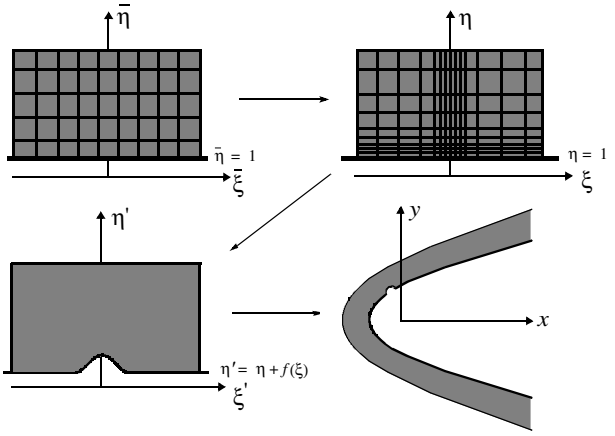


Fig. 3. Schematic diagram of the overall transformation from the computational space to physical space.

and the solution tends towards the inviscid value, which is known analytically for the parabola). Using this technique, the total streamfunction, ψ, and total vorticity, ω, are given by

$$\psi = \Psi + \psi_{\text{inv}} \text{ and } \omega = \Omega + \omega_{\text{inv}}. \tag{5}$$

ψ_{inv} and ω_{inv} are the inviscid solutions, where $\omega_{\text{inv}} = 0$ for potential flow. The terms Ψ and Ω are the viscous solutions. Applying the transformation equations for parabolic coordinates, surface roughness, grid generation, and implementing the flow variable splitting technique, Eqs. (1) and (2) can be written in the final form:

$$\begin{aligned}&\bar{\xi}_{\xi\xi}\Psi_{\bar{\xi}} + (\bar{\xi}_{\xi})^2\Psi_{\bar{\xi}\bar{\xi}} - f_{\xi\xi}\bar{\eta}_{\eta}\Psi_{\bar{\eta}} - 2f_{\xi}\bar{\xi}_{\xi}\bar{\eta}_{\eta}\Psi_{\bar{\xi}\bar{\eta}} + (1+(f_{\xi})^2)\bar{\eta}_{\eta\eta}\Psi_{\bar{\eta}} + (1+(f_{\xi})^2)(\bar{\eta}_{\eta})^2\Psi_{\bar{\eta}\bar{\eta}}\\ &\quad = (-h^2)\Omega + \Psi_{\tau}\end{aligned} \tag{6}$$

and

$$\begin{aligned}&\Omega_{\tau} + h^2\Omega_t + \bar{\xi}_{\xi}\bar{\eta}_{\eta}(\Psi_{\bar{\eta}}\Omega_{\bar{\xi}} - \Psi_{\bar{\xi}}\Omega_{\bar{\eta}}) + (\psi_{\text{inv}})_{\eta}\bar{\xi}_{\xi}\Omega_{\bar{\xi}} - f_{\xi}(\psi_{\text{inv}})_{\eta}\bar{\eta}_{\eta}\Omega_{\bar{\eta}} - (\psi_{\text{inv}})_{\xi}\bar{\eta}_{\eta}\Omega_{\bar{\eta}}\\ &\quad = Re^{-1}\left[\bar{\xi}_{\xi\xi}\Omega_{\bar{\xi}} + (\bar{\xi}_{\xi})^2\Omega_{\bar{\xi}\bar{\xi}} - f_{\xi\xi}\bar{\eta}_{\eta}\Omega_{\bar{\eta}} - 2f_{\xi}\bar{\xi}_{\xi}\bar{\eta}_{\eta}\Omega_{\bar{\xi}\bar{\eta}} + (1(f_{\xi})^2)\bar{\eta}_{\eta\eta}\Omega_{\bar{\eta}} + (1+(f_{\xi})^2)(\bar{\eta}_{\eta})^2\Omega_{\bar{\eta}\bar{\eta}}\right],\end{aligned} \tag{7}$$

where $h^2 = \xi^2 + \eta^2 + 2\eta f + f^2$. To help stabilize the method during global iterations, a fictitious time-derivative term, or pseudo-time term, is added to both the streamfunction Poisson equation (6) and the vorticity transport equation (7) (i.e. Ψ_{τ} and Ω_{τ}, respectively). At each physical time step, the solution is converged in pseudo-time prior to moving to the next time step.

2.1. Inviscid flow

There exists an analytic solution for the inviscid potential flow past a parabola from Werle and Davis [27] and Van Dyke [34]. For a clean parabola, the inviscid streamfunction is given by

$$\psi_{\text{inv}} = (\xi' + K)(\eta' - 1), \tag{8}$$

where K is the vertical displacement of the stagnation point from the vertex of the parabola and is termed the angle-of-attack parameter. K can be related to the angle-of-attack of the airfoil when the parabolic leading edge is coupled to a thin airfoil description of the full airfoil. For this work, the parabola is set to a 0° angle-of-attack, which translates to $K = 0.0$. Applying the Prandtl transposition to Eq. (8) produces the inviscid potential solution for the flow past a parabola with surface roughness present:

$$\psi_{\text{inv}} = (\xi + K)(\eta + f(\xi) - 1). \tag{9}$$

2.2. Initial conditions

For the simulations presented in this work, the parabola used an impulsively started flow. The effects of additional freestream perturbations on the velocity fluctuations and the unsteady vortex-shedding mechanism were also investigated. There were two different freestream perturbations used: (a) continuous oscillation of the parabola leading edge and (b) the addition of inviscid

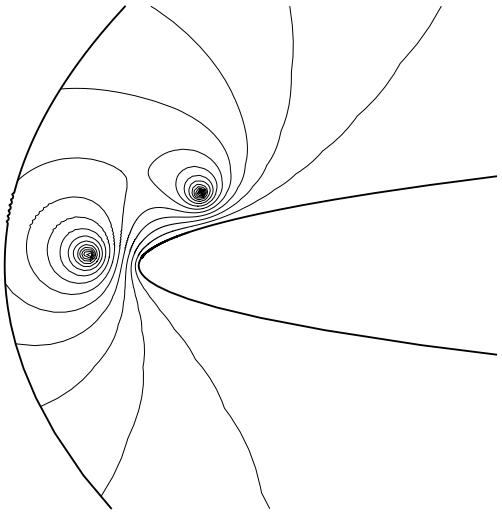


Fig. 4. Streamfunction contours of the inviscid vortices inserted as a freestream perturbation.

freestream vortices. To generate the oscillations of the leading edge, a sinusoidal variation of $K(t)$ was implemented as

$$K(t) = K_{\text{mean}} + K_{\text{amp}} \sin\left(\frac{2\pi t}{\lambda}\right). \tag{10}$$

The parabola was set to oscillate about the zero degree angle-of-attack position, so K_{mean} was set to 0.0, while the period was set to $\lambda = 2$.

Possibly a more natural mechanism for the generation of freestream disturbances is the addition of inviscid vortices. The vortex strength was set so that the perturbation to the velocity field was less than 1%. Two vortices were used to generate the freestream perturbation, which is shown in Fig. 4. Note that the figure is only displaying the streamfunction contours of the inviscid vortices at their initial position. The vortices were then convected downstream in physical time. As a first approximation, the vortices are convected along the streamlines of the main flow. The impact of these added perturbations is analyzed by comparing results to the case with no freestream perturbations.

2.3. Boundary conditions

No-slip boundary conditions, with $u = v = 0$, are applied at the surface of the parabola ($\eta = 1$). In terms of the streamfunction, this boundary condition is $\psi = 0$ and $\psi_\eta = 0$. Applying the flow-variable splitting, these boundary conditions become

$$\Psi = -\psi_{\text{inv}} \text{ and } \Psi_{\bar{\eta}} = -\frac{(\psi_{\text{inv}})_\eta}{\bar{\eta}_\eta} \text{ at } \eta = 1. \tag{11}$$

At the far-field boundary, with η large but finite, the viscous effects die out and a fully inviscid flow is recovered. Therefore, the viscous part of the split variables, Ψ and Ω, is assumed to be zero at this boundary. The resulting far-field boundary conditions are

$$\psi = \psi_{\text{inv}} \text{ and } \Omega = 0 \text{ as } \eta \to \infty. \tag{12}$$

The viscous solution for flow past a parabola is known to approach the Blasius solution as $\xi \to \infty$ (Davis [28]). Therefore, at the far-downstream boundary, the flow is assumed to be steady and the expressions from Davis [28] for the Blasius solution are used to evaluate the streamwise derivatives. The upper and lower downstream boundaries are placed at a large, but finite values of $|\xi|$.

2.4. Unsteady flow parameters

In this study a comparison is made between the unsteady flow results from the current algorithm and the roughness-induced transitional flow from the experimental work by Bragg et al. [12]. One item that is used as a comparison device is the measured turbulence intensity values from the experiment. The convention that will be used here is to label these types of values calculated from the current algorithm as "fluctuation intensity" values, while the experimental results will retain the standard label of turbulence intensity. The turbulence intensity is defined to be the root-mean square of the perturbation velocity. The algorithm uses the same equation as the experimental work to generate the fluctuation intensity values, as described below.

In this section, we only focus on the u-component as a representative example, but the same procedure is equally valid for either the total velocity or any of the velocity components. Using the fact that the total u velocity component is equal to the mean plus the perturbation component, the following equation may be written for u^2:

$$u^2 = \bar{u}^2 + 2\bar{u}u' + u'^2. \tag{13}$$

If the unsteady simulation is run out to a sufficiently long time, then the mean velocity, $\bar{u}$, will be constant everywhere in the flowfield. Assuming this has been achieved and using the assumption that $\langle u' \rangle = 0$, then taking the time-average of Eq. (13) gives the following:

$$\langle u^2 \rangle = \bar{u}^2 + \langle u'^2 \rangle. \tag{14}$$

The turbulence intensity is found by taking the square-root of the time-averaged square of the perturbation velocity at the completion of the unsteady run, i.e.

$$u'_{\text{rms}} = \left[\langle u'^2 \rangle \right]^{1/2}. \tag{15}$$

The use of this method allows the calculation to be completed in a single run; the $\bar{u}$ values do not have to be calculated prior to using this equation, assuming that the unsteady run has proceeded long enough in time that $\bar{u}$ and u'_{rms} are constant everywhere. The total velocity or other velocity components can be calculated in a similar fashion.

To compute the fluctuation (or turbulence) intensity given above, the velocity components must first be extracted from the streamfunction generated by the flow solver. Using the standard definition of the two-dimensional streamfunction in Cartesian coordinates and applying appropriate coordinate transformations to these equations, the velocity components are found to be

$$u = \beta \Big\{ (\eta + f)(\bar{\xi}_{\xi} \psi_{\bar{\xi}} - f_{\xi} \bar{\eta}_{\eta} \psi_{\bar{\eta}}) + \xi \bar{\eta}_{\eta} \psi_{\bar{\eta}} \Big\}, \tag{16}$$

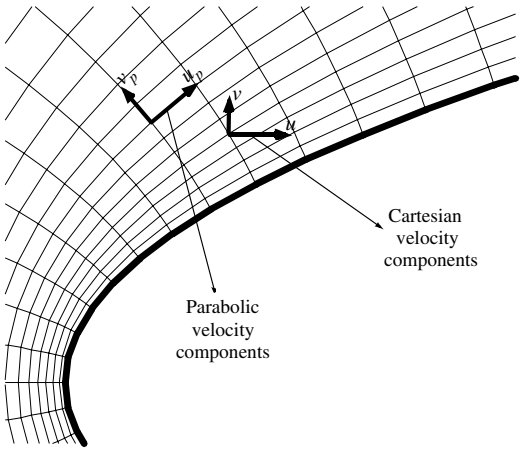


Fig. 5. The velocity components in the Cartesian and parabolic coordinate systems.

$$v = \beta\Big\{-\xi(\bar{\xi}_{\xi}\psi_{\xi} - f_{\xi}\bar{\eta}_{\eta}\psi_{\eta}) + (\eta + f)\bar{\eta}_{\eta}\psi_{\eta}\Big\}, \tag{17}$$

where

$$\beta = \frac{1}{(\xi^2 + \eta^2 + 2\eta f + f^2)}. \tag{18}$$

These equations give the u and v velocity components in the Cartesian coordinate system, as shown in Fig. 5. This is required to make the proper comparison to the experimental transition work. The velocity profile measurements reported by Bragg et al. [20] were made going out from the airfoil surface, perpendicular to the tunnel axis. For this reason, the velocity components need to be aligned with the Cartesian coordinate system, and not the parabolic, body-fitted coordinates.

In addition, the experimental results for the turbulence intensity were obtained with a single hot-wire probe, which gives the total velocity (two-dimensional projection) at a given boundary-layer location. Therefore, the current algorithm calculates fluctuation intensity values from the total two-dimensional velocity, given by

$$u'_{\mathrm{rms}} = \left[\langle u'^2\rangle + \langle v'^2\rangle\right]^{1/2}. \tag{19}$$

This allows for the proper comparisons to be made to the experimental work [20].

2.5. Numerical method

An implicit numerical method is used to solve the streamfunction and vorticity transport equations. The numerical scheme is second-order accurate in time and space, with central differences used for the spatial derivatives and a backward difference for the temporal derivative. The non-linear convective terms are linearized using Newton linearization, while mixed-derivatives are treated iteratively as source terms.

As a starting solution for the full Navier–Stokes algorithm, the parabolized Navier–Stokes equations are solved for flow past a parabola at a constant angle-of-attack and the solution is used as input for the full Navier–Stokes calculations. The parabolized Navier–Stokes solver uses the Blasius solution as a starting solution.

The implicit scheme sweeps in both the ξ and η directions, with the sweeps in the ξ direction being bi-directional and the sweeps in η being uni-directional. One complete spatial sweep consists of two alternating sweeps in the streamwise direction (ξ) and one sweep in the normal direction (η), which is equivalent to one pseudo-time step. Each sweep involves the solution of a 2×2 block-tridiagonal system of equations.

For unsteady flow, the code marches the solution in time and at each physical time step, the pseudo-time terms are iterated to convergence before moving to the next physical time step. The algorithm has the capability of solving for purely steady flow by either: (1) allowing the unsteady code to eventually converge to a steady solution; or (2) eliminating the time term, Ω_t, and solving the remaining steady Navier–Stokes equations. With this particular algorithm, if the flow field is truly steady, then either method can be used to obtain the solution. If the flow is unsteady, then the "steady" flow solver may not converge.

For a majority of the computations included in this study, the non-dimensional physical time step was 1×10^{-2}. Using the NACA 0012 airfoil at a chord Reynolds number of 1.25×10^{6} as a reference, the Δt for the parabola translates to a dimensional time step of 2.5×10^{-6} s. It should be noted that the time step was varied during testing to ensure a time-independent solution. The time step was reduced until the resulting $\bar{u}$ and fluctuation intensity values were unchanged for given flow condition.

The grid used for the computations is a structured, body-fitted grid. A region of dense, uniformly spaced grid points in the streamwise direction is placed in the vicinity of the parabola leading edge, with hyperbolic stretching upstream and downstream of this region. Stretching is also used in the normal direction to cluster points near the surface. A typical parabolic grid is shown in Fig. 6. The grid is 481×81, with 380 streamwise grid points placed in the dense-grid region near the leading edge. In this figure, the leading edge of the parabola has a clean surface (i.e. no roughness). When roughness is present, typically the grid requirements are increased to properly resolve the flow about each roughness element. In this study, the grids ranged from 881×81 to 1501×201. In general, the flow solution was more sensitive to the grid density in the streamwise direction than in the normal direction when surface roughness was present, assuming that adequate grid resolution was used in the normal direction for the corresponding clean surface.

2.6. Surface protuberance

The surface protuberance is added to the baseline parabolic surface using a Prandtl transposition which allows new surface oriented coordinates to be easily generated without the need for re-gridding (i.e. alterations to the surface roughness geometry can be made without changing the actual grid parameters).

The current work uses an analytic roughness geometry for the surface protuberance, which produces a smooth hump with variable height and wavelength. The hump equation used is given by

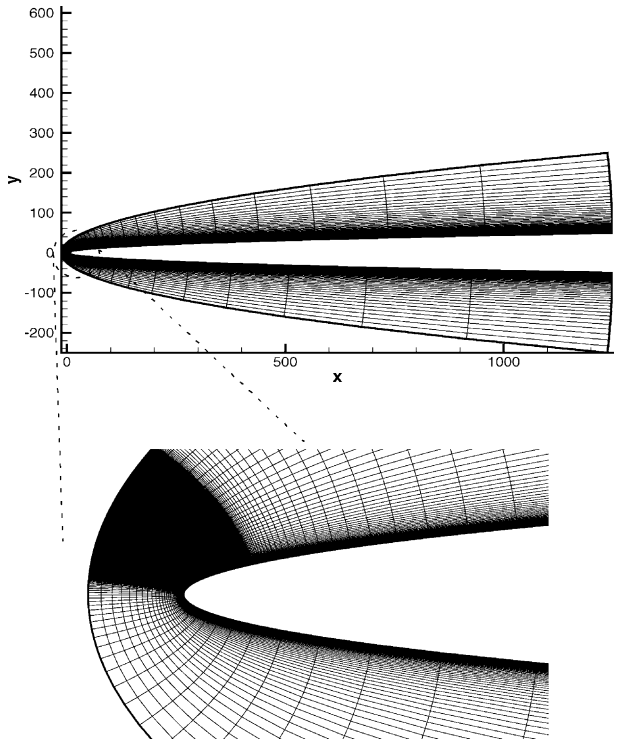


Fig. 6. Grid used for flow past a parabola with a dense-grid region placed on the leading edge.

$$f(\xi) = h(\xi - \xi_{\mathrm{L}})^5(\xi_{\mathrm{R}} - \xi)^5, \tag{20}$$

where h is the hump height and ξ_L, ξ_R are the left and right endpoints of the hump, respectively. A 10th-order polynomial was used to ensure a smooth transition from the roughness element to the parabola surface. This helped to alleviate any curvature discontinuity problems at the juncture point.

In this study, only simple geometrical leading-edge roughness is investigated, but more complex roughness can be incorporated onto the parabola surface by superimposing multiple roughness elements of varying heights and wavelengths. (see [26]). The simple roughness is either a single smooth hump placed on the parabola leading edge or multiple humps with equal height and wavelength arranged to simulate distributed roughness.

2.7. Parabola conversion

One primary objective of this study was to compare results from the current Navier–Stokes algorithm to available experimental results. However, the current results are for flow past a parabola, while the experimental results are from an actual airfoil. Therefore, a necessary part of the analysis is the conversion of common parameters, which include the Reynolds number and certain geometrical characteristics.

The characteristic length-scale for flow past the parabola is the leading-edge radius of curvature. The Reynold's number used for the parabola (Re_l) is based on this leading-edge radius. (Re_l) is related to the airfoil chord Reynolds number by

$$Re_l = \frac{l}{c} Re_c, \tag{21}$$

where l and c are the leading-edge radius of curvature of the parabola and the chord length of the airfoil, respectively. For the NACA 0012 airfoil, the (l/c) ratio is equal to 0.016 [35]. This allows computation of the flow in a localized region at a significantly lower Reynolds number, Re_l, than that of the chord Reynolds number (approximately two orders-of-magnitude). For the majority of cases presented here, Re_l varied from 1000 to 20,000. For the NACA 0012 airfoil, the corresponding Re_c ranges from 0.63×10^5 to 1.25×10^6.

The geometry conversions mainly involve transferring the airfoil surface roughness to the parabola surface. This is accomplished by first dividing the arc length or height values of the roughness by the airfoil chord length. This non-dimensional value is divided by the same (l/c) ratio used in the Reynolds number conversion. A shift in the x-coordinate is then applied since the leading edge of the parabola is located at $x = -0.5$. Therefore, the conversion is given by

$$X_{\text{parabola}} = \frac{(x/c)_{\text{airfoil}}}{(l/c)} - 0.5. \tag{22}$$

This produces an accurate representation of the airfoil roughness geometry on the parabola surface.

3. Comparisons to transitional flow work

The primary goal of this work was to investigate unsteady flow, velocity fluctuations, and vortex shedding past large-scale roughness on the leading-edge of an airfoil, while making comparisons to experimental roughness-induced transition results. However, some preliminary work was also done for steady flow past simple roughness on the leading edge of the parabola. A portion of these results are presented here as a reference for steady flow solutions. This algorithm has been tested previously to verify the validity of the results. These validation results can be found in the work by Huebsch and Rothmayer [26].

Several different cases were run with the simple roughness geometries on the leading-edge of the parabola. The first case was a single hump on the upper leading-edge surface, with the hump being fully contained within the boundary layer; the hump height was approximately 1/3 of the approaching laminar boundary layer height. A streamfunction contour plot is shown in Fig. 7. This particular case was at a constant angle-of-attack of $K = 0.0$ (0° angle-of-attack), $Re_l = 1000$, and a grid of 881×81, with the dense-grid boundaries placed upstream and downstream of the hump. Both the steady and unsteady flow solvers were applied to this case, and both produced identical results. The unsteady flow solver converged to a steady solution. As the figure shows, the hump geometry produced a steady, laminar separation bubble on the downstream-side of the hump. Vorticity contours have also been plotted for this case and are shown in Fig. 8. Qualitatively, this

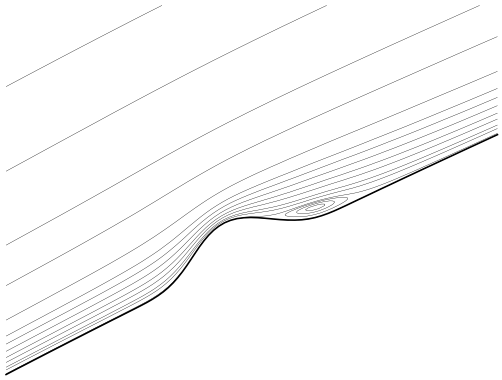


Fig. 7. Streamfunction contours for flow past a single analytic hump placed on the leading edge showing a steady laminar separation bubble ($Re_l = 1000$).

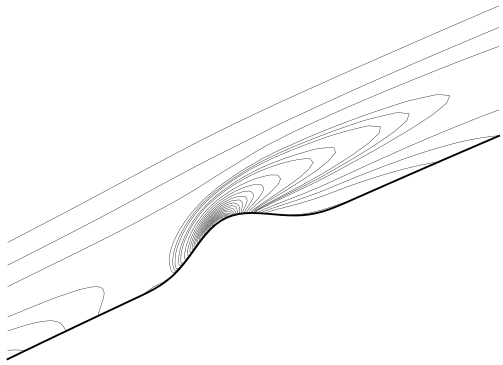


Fig. 8. Vorticity contours for flow past a single analytic hump.

plot is consistent with results for flow past a hump on a flat plate, such as in the work done by Kiya and Arie [36] and Haussling [37].

Grid independence of the flow solution was periodically checked throughout the course of this work. For the single leading-edge roughness element, results for the grid dependence test are presented in Fig. 9, which shows that the grid resolution used in this case is sufficient to capture the flow past the isolated hump. For a grid independent solution, the required grid resolution is approximately 40–50 streamwise grid points for an individual hump. As expected, the evaluation of grid dependence in this work revealed that increases in roughness complexity and flow gradients require a finer grid to resolve the flow physics correctly.

A second case of simple roughness at a moderate Reynolds number was tested, which consisted of eight simple humps placed on the upper-surface leading edge to simulate distributed roughness. The flow conditions were the same as for the single hump case: a constant angle-of-attack of $K = 0.0$ and $Re_l = 1000$. The unsteady flow solver converged to a steady-state solution for this roughness geometry. As in the single hump case, the distributed roughness was fully contained within the boundary layer. The dense-grid region was fine enough to capture the small-scale

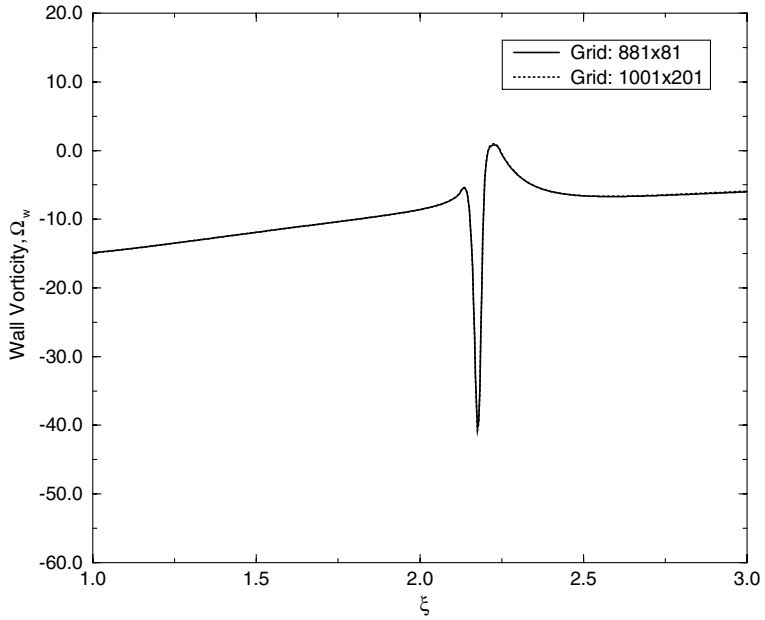


Fig. 9. Grid dependence study for flow past a single hump on the leading edge, shown in Fig. 7.

separation regions contained between the roughness elements. Fig. 10 shows the streamfunction contour plot for this case. It also includes a close-up showing the local separation between the roughness elements. This case produced steady, laminar flow separation for the given Re_1.

The main focus of this investigation was to predict unsteady flow parameters, such as the fluctuation intensity, and compare these to experimental transitional flow results. The primary work for comparison comes from experiments performed by Bragg et al. [20], where they investigate transitional flow on a NACA 0012 airfoil with large leading-edge roughness that is characteristic of early glaze ice growth.

3.1. Fluctuation intensity comparison

In the work by Bragg et al. [20], several different distributed roughness geometries were used, along with several isolated roughness elements. The distributed roughness consisted of hemispheres placed in staggered rows with a height of 0.35 mm. The geometrical differences in the distributed roughness cases are found in the chordwise extent of the roughness (the roughness patch length) and the arc length from the leading edge to the location where the roughness begins.

For our study, the 1/4 in. roughness case [20] was selected for comparison. The 1/4 in. label refers to the chordwise extent of the roughness patch. A schematic of the distributed roughness geometry is given in Fig. 11. The start of the roughness patch is at a surface arc length of 7 mm from the leading edge, while the aft edge of the roughness is at a surface arc length of 13.25 mm. The spacing of the roughness elements (center-to-center) is 1.3 mm. It should be noted that the work by Bragg et al. [20] uses three-dimensional hemispheres, whereas our roughness elements are two-dimensional humps. The reference airfoil used in the experimental work is a NACA 0012 with

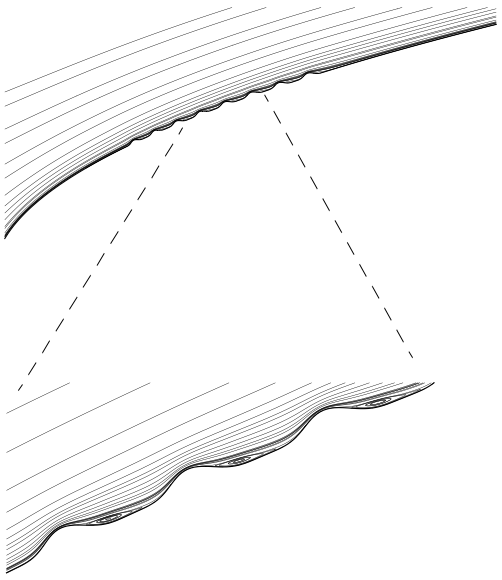


Fig. 10. Streamfunction plot for simple roughness (8 hump) with steady, laminar flow at 0° angle-of-attack ($Re_l = 1000$).

a chord length of 21 in. The roughness geometry shown in Fig. 11 is the actual airfoil roughness incorporated onto the leading edge of the parabola, with the corresponding airfoil measurements given as reference.

Fig. 11 also includes a picture of a typical grid that is used for the calculation of the unsteady flow parameters. The grid shown is 1001×201, with 900 of the streamwise grid points located in the dense-grid region on the upper leading-edge surface. The 1/4 in. roughness is enclosed within this fine-grid region, which also extends downstream to approximately 7% of chord. In terms of the airfoil chord, the distributed roughness region is contained within $0.00490 \leqslant x/c \leqslant 0.0138$. During the course of this work, the grid density was increased and the extent of the fine-grid region was varied to ensure grid independent solutions.

In addition to the various roughness geometries, Bragg et al. [20] used three different chord Reynolds numbers in their study. In the present work, comparisons are made to the first two values: $Re_c = 0.75 \times 10^6$ and $Re_c = 1.25 \times 10^6$. Unsteady, viscous flow simulations for the parabola at 0° angle-of-attack with the 1/4 in. roughness were conducted for the two chord Reynolds numbers given above. The results of these simulations are shown in Fig. 12, which is a plot of the instantaneous streamfunction. In Fig. 12a, the flow in the roughness region is steady, laminar flow for $Re_c = 0.75 \times 10^6$. When the chord Reynolds number is increased to $Re_c = 1.25 \times 10^6$, the flow becomes fully unsteady, with vortex shedding. Fig. 12b shows the bubbles being convected downstream at the aft edge of the roughness. The structure of the vortex shedding is similar to results reported by Pauley et al. [38] for flow separation and vortex shedding on a flat plate. Streamline results from Pauley et al. [38] are shown in Fig. 13, which, qualitatively, compare well with the streamlines in Fig. 12b. Fig. 14 shows a more physically realistic picture of the vortex shedding process downstream of the roughness by plotting the vorticity contours for the

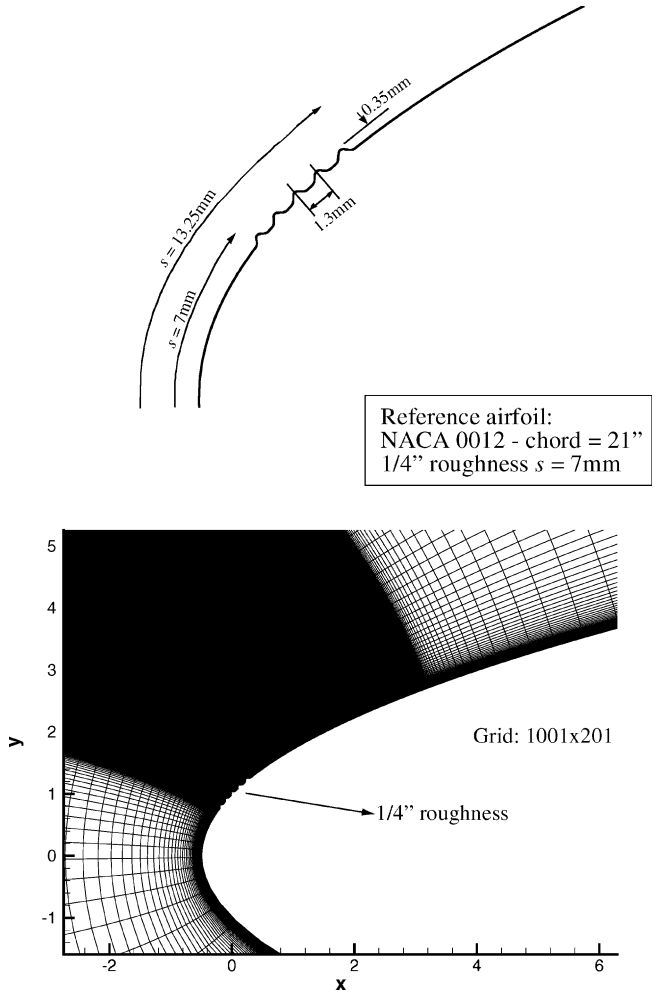


Fig. 11. Schematic diagram of Bragg et al.'s [20] 1/4 in. roughness applied to the parabola surface. Also shown is a typical grid used for the unsteady, vortex-shedding—1001×201.

$Re_c = 1.25 \times 10^6$ case. Fasel et al. [16] postulates that surface roughness producing this type of vortex shedding actually acts as a source of additional disturbances and augments the inherent natural disturbances that occur, which can promote early transition.

To provide a sense of the grid resolution used in this type of simulation, Fig. 15 shows a close-up of the streamfunction contours for the last roughness element along with the associated grid. From this figure, it can be seen that the grid is quite adequate to capture the vortex-shedding phenomenon. For reference, if the patched grid region of the parabola were placed on the NACA 0012 airfoil, the streamwise grid spacing would be 5.3×10^{-5} with respect to x/c, or approximately 0.005% of chord. In terms of the normal direction on the airfoil, the first grid point from the wall is approximately one order of magnitude less than the streamwise direction (7.1×10^{-6}). Of course the grid in the normal direction is stretched so the spacing will increase as you move away from

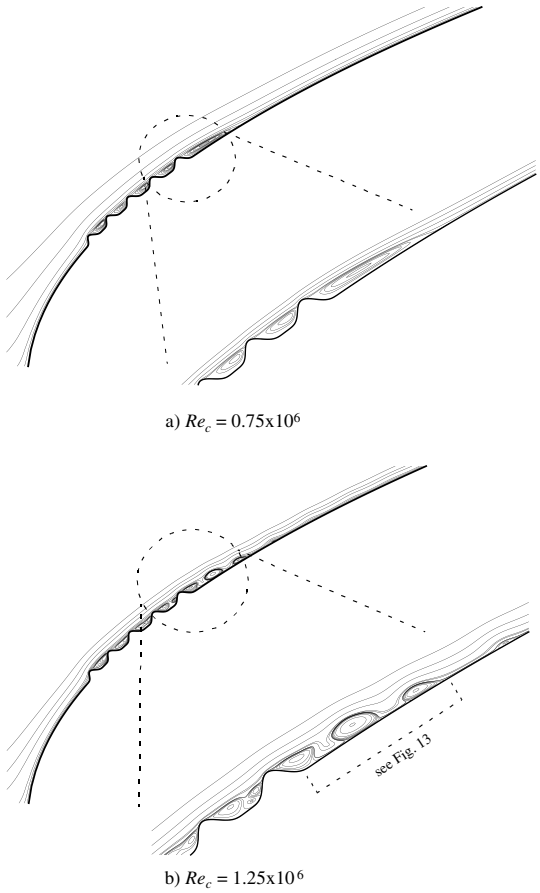


Fig. 12. Comparison of 1/4 in. roughness at different Re_c values (streamfunction contours).

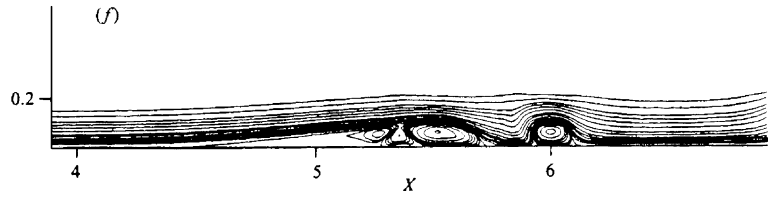


Fig. 13. Instantaneous streamline results from Pauley et al. [38] for two-dimensional separation on a flat plate with vortex shedding.

the surface. As a rough estimate of how well the present solutions might capture smaller scale structures, consider a size-ratio of boundary-layer scaled vorticies to a smaller scale, such as $(Re^{-1})/(Re^{-1/2})$, which is 7×10^{-3} for the solution shown in Fig. 15. The ratio of the smallest locally uniform grid spacing used within the boundary layer to a typical boundary-layer scaled vortex size is roughly 5×10^{-2}. Therefore, it is evident that while the grid resolution used in this

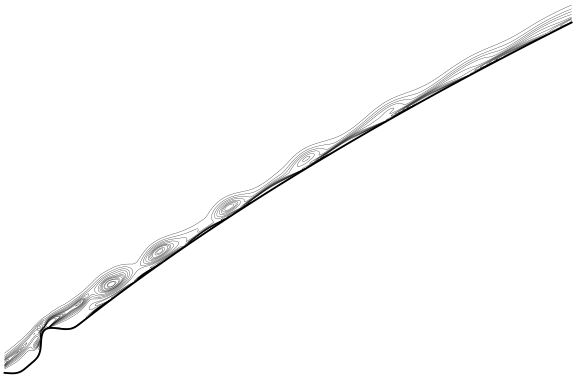


Fig. 14. Vortex shedding for the 1/4 in. roughness at $Re_c = 1.25 \times 10^6$ (vorticity contour plot).

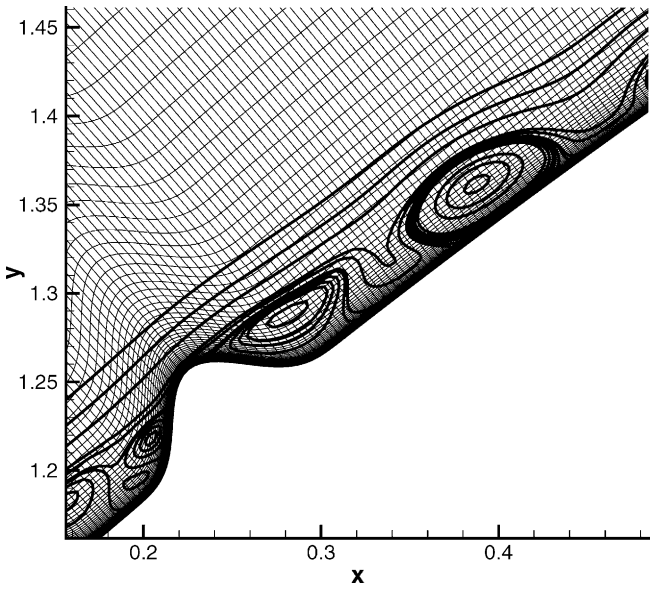


Fig. 15. Close-up of the last roughness element showing the typical grid resolution used to capture the vortex shedding process.

study adequately captures the structures computed in Fig. 15, the authors anticipate that it will not resolve scales near (Re^{-1}). Other typical Reynolds number scaling ratios include: $(Re^{-3/4})/(Re^{-1/2}) = 8 \times 10^{-2}$ and $(Re^{-1}\ln(Re))/(Re^{-1/2}) = 7 \times 10^{-2}$. Obviously these scale differences will become worse as the chord Reynolds number is increased.

Bragg et al. [20] summarize the boundary-layer state for the clean airfoil and all distributed roughness cases in a chart showing regions of laminar, transitional, and turbulent flow along the airfoil chord. The current work has investigated two of these cases for flow in the leading-edge region: the clean airfoil case and the 1/4 in. distributed roughness case at two different Reynolds numbers. The flow characteristics from this work are summarized in the chart below.

$Re_c = 0.75 \times 10^6$	Clean model	Laminar, steady flow
	1/4 in. s = 7 mm	Laminar, steady flow
$Re_c = 1.25 \times 10^6$	Clean model	Laminar, steady flow
	1/4 in. s = 7 mm	Unsteady, vortex shedding

These results show excellent agreement with the experimental results. The chart of Bragg et al. [20] shows laminar flow regions near the leading edge for the three laminar flow cases cited above. We concluded that these three cases produced laminar, steady flow based on several observations:

(1) the converged solution shows steady, laminar separation behind and downstream of the roughness;
(2) the calculated fluctuation intensity values are zero everywhere in the flow field;
(3) the code converges to a steady solution for these cases using both a steady code and an unsteady code.

For the case with the distributed roughness at the higher chord Reynolds number, Bragg et al. [20] found transitional flow in the leading edge region. Our simulation shows there is no laminar, steady flow, but rather the flow is fully unsteady with vortex shedding at the aft edge of the roughness. In addition, the calculated fluctuation intensity values are non-zero downstream of the distributed roughness. There may be some possible correlation between these roughness-induced flow phenomena and the onset of transitional flow when surface roughness is present.

The fluctuation intensity contours for the 1/4 in. roughness at $Re_c = 1.25 \times 10^6$ are shown in Fig. 16. Clearly, the fluctuation intensity is non-zero downstream of the roughness patch, with the peak u'_{rms} values a short distance downstream of the last roughness element. The peak intensity value is approximately 17%.

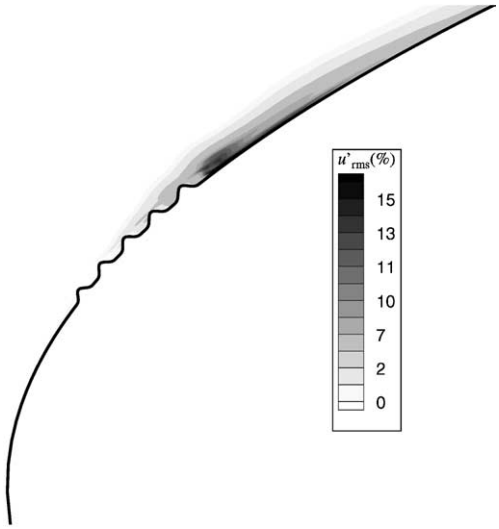


Fig. 16. Turbulence intensity (%) contours for the 1/4 in. roughness model at $Re_c = 1.25 \times 10^6$.

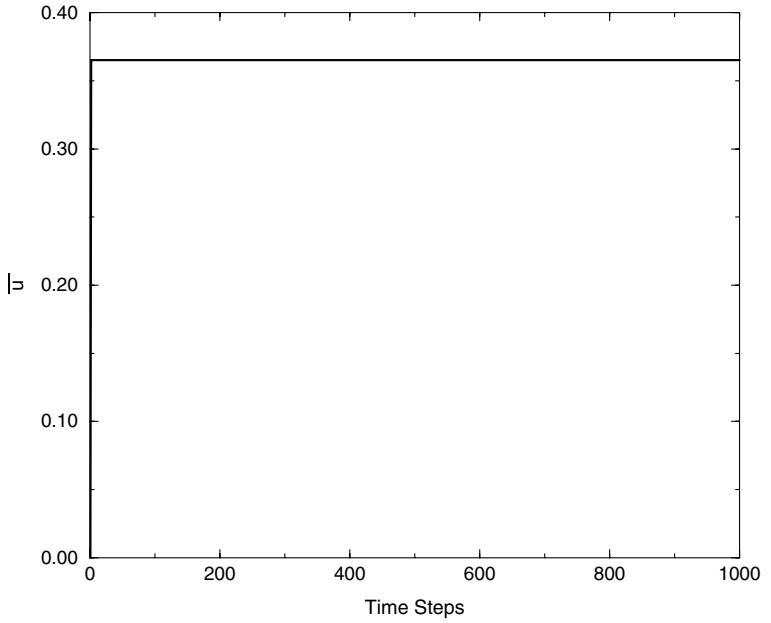


Fig. 17. Time history of the mean velocity during an unsteady run with the 1/4 in. roughness at $Re_c = 0.75 \times 10^6$.

As stated previously, one of the assumptions in the calculation of u'_{rms} is that the simulation is run for a long enough time so that the value of $\bar{u}$ (and u'_{rms}) becomes constant everywhere in the flow field. This is a necessary condition for the analysis method to hold true. Examples of this are shown in Figs. 17 and 18. During an unsteady run, $\bar{u}$ was monitored at several locations in the flow field, with one of these locations downstream of the roughness near the surface where high velocity fluctuations were anticipated. Fig. 17 shows the results of monitoring $\bar{u}$ for the 1/4 in. roughness case at $Re_c = 0.75 \times 10^6$, which is nearly constant throughout the unsteady run. For this case, the flow was steady and laminar, which is consistent with little or no fluctuation in the velocity at a given location.

The value of $\bar{u}$ was also monitored for the higher chord Reynolds number case ($Re_c = 1.25 \times 10^6$) and one of the results is given in Fig. 18. This case produces more changes in the time-averaged u-velocity values, and after 3000 time steps, the $\bar{u}$ value has not yet reached a constant. For this particular case, $\bar{u}$ leveled off at approximately 7000 time steps. For reference, an unsteady run of 7000 physical time steps and a grid of 1001×101 took approximately three days on a 600 MHz DEC alpha workstation with one gigabyte of RAM.

A time history of a turbulence intensity profile at a given location downstream of the 1/4 in. roughness is provided in Fig. 19. During the unsteady run, the u'_{rms} values eventually converge to a time-independent value, which in this case is at $t = 70$. The plot also includes the u'_{rms} profile for $t = 80$, which is unchanged from the $t = 70$ profile. Therefore, the u'_{rms} and the $\bar{u}$ values eventually converge to a constant value.

Bragg et al. [20] also provide turbulence intensity profiles for various chord locations downstream of the distributed roughness. Fig. 20 shows the experimental results [20] for the NACA 0012 with the 1/4 in. roughness at $Re_c = 1.25 \times 10^6$ for $x/c = 0.05$. The plot also includes the

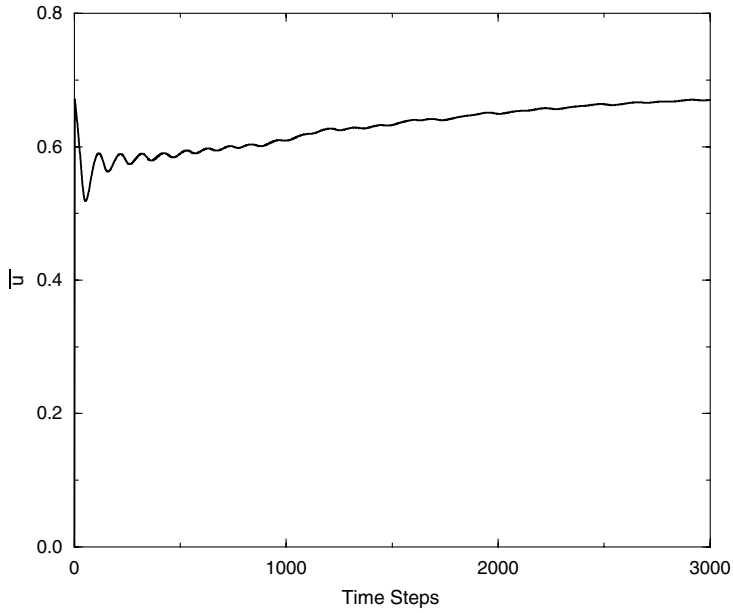


Fig. 18. Time history of the mean velocity during an unsteady run with the 1/4 in. roughness at $Re_c = 1.25 \times 10^6$.

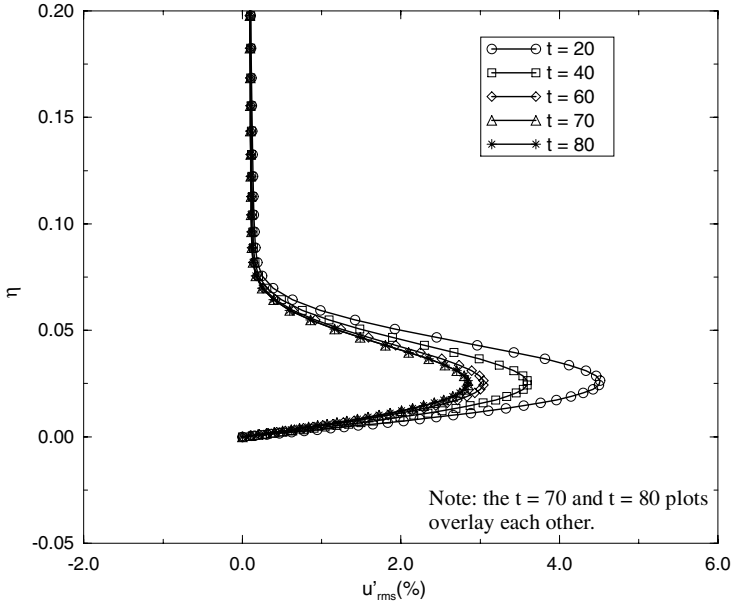


Fig. 19. A time history of turbulence intensity profiles at $X_{\text{parabola}} = 3.14$ ($x/c = 0.05$) for 1/4 in. roughness with $Re_c = 1.25 \times 10^6$.

fluctuation intensity profile for the corresponding location on the parabola leading edge—$X_{\text{parabola}} = 3.14$. The results from the parabola were obtained with the same distributed roughness

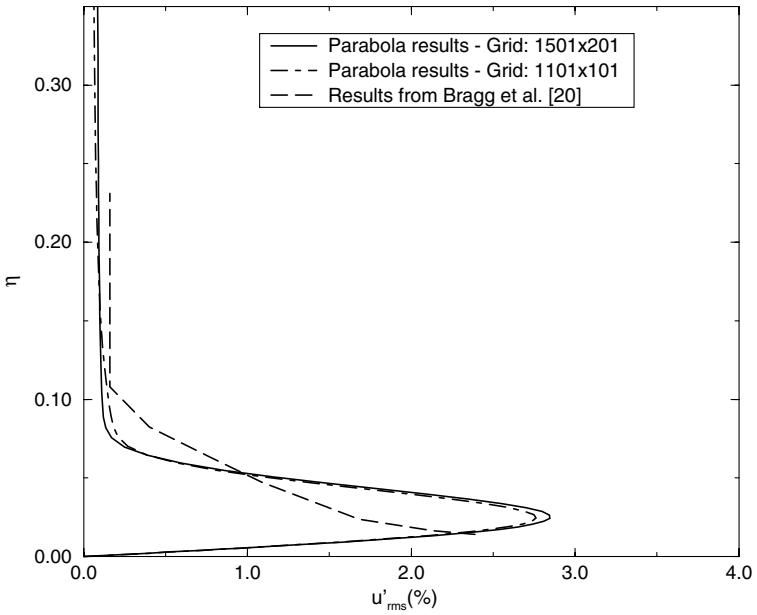


Fig. 20. A comparison of fluctuation intensity profiles to a turbulence intensity profile from the experimental work of Bragg et al. [20] for the 1/4 in. roughness at $x/c = 0.05$.

geometry and the same chord Reynolds number as that of the experimental data. This plot shows that the overall magnitude and trends of the two-dimensional Navier–Stokes results are in good agreement with the experimental results. Any discrepancies could be caused by three-dimensional effects. The fluctuation intensity results provide a reasonable match to the experimental turbulence intensity for the roughness geometry considered here.

Fig. 20 includes fluctuation intensity profiles from the parabola for two different grids, 1501×201 and 1101×101, which are in good agreement. The downstream boundary of the dense-grid region was also varied to ensure that the placement did not effect the upstream flow and the profile measurements. For comparison, if the 1501×201 parabola grid is extrapolated out to the full NACA 0012 airfoil, the corresponding grid resolution would require approximately 21,000 streamwise grid points to match the resolution of the dense-grid region on the leading edge of the parabola.

During the grid dependence testing for the 1/4 in. roughness case, it was found that if the dense-grid region did not have the proper resolution, the unsteady solver would converge to a steady solution, even at the highest chord Reynolds number. Therefore, a lack of grid resolution can have a significant effect on this type of unsteady flow calculations. The coarse grid produces a strong dissipative effect that cannot maintain the unsteady flow, thus giving a "pseudo-steady" solution.

It is well known [39] that turbulence intensity will have a peak value near the surface, then decrease to zero at the wall. The fluctuation intensity profiles from Fig. 20 demonstrate this same trend. The experimental turbulence intensity curve does not bend back to zero, but it was not clear exactly how close to the airfoil surface the data was taken. Referring to the work by Winkler [21], it is noted that data was obtained within 0.5 roughness heights of the surface due to the finite size

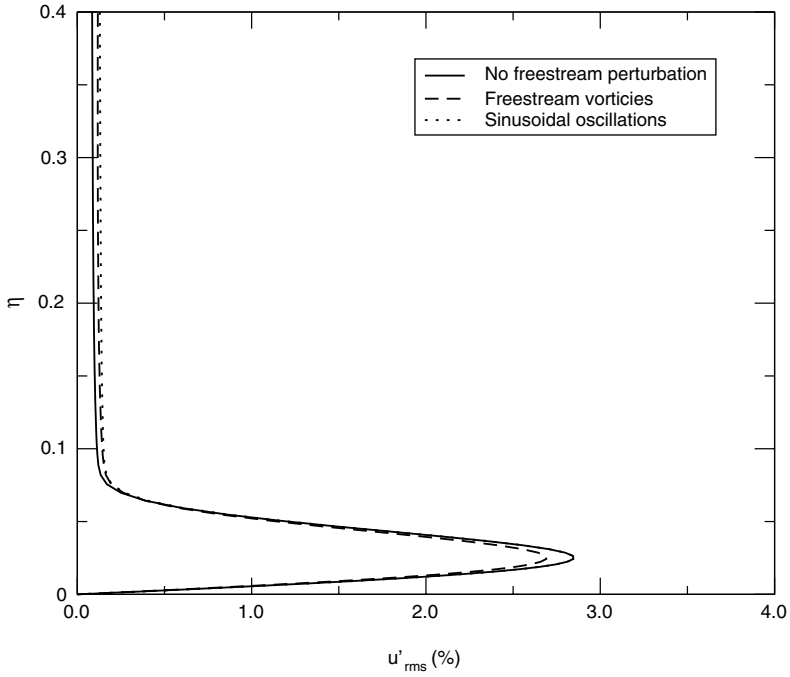


Fig. 21. Freestream perturbation effects on fluctuation intensity.

of the hot-wire probe. With a roughness height of 0.35 mm, the closest data point should be at approximately 0.175 mm from the surface. The peak fluctuation intensity value on the parabola at the $x/c = 0.05$ station would correspond to a distance of approximately 0.2 mm from the surface of the NACA 0012 in the Bragg et al. [20] experiment.

As discussed in the section on initial conditions, the effects of freestream perturbations were also investigated with respect to velocity fluctuations and vortex shedding. The case shown in Fig. 20 was repeated with both the leading-edge oscillation and the freestream vortices. The grid of 1501×201 was used for these simulations, which were run out to the same physical time. The fluctuation intensity profiles for the two freestream perturbation cases are shown in Fig. 21 and are compared to the case with no freestream perturbation (i.e. Fig. 20). These results show that the u'_{rms} profiles are quite similar for the no-disturbance case and the two freestream perturbation cases. As expected, at early times, there were some differences in the results. However, at $t = 70$, the results compare fairly well. Therefore, the unsteady forcing had minimal impact at longer times. This suggests that large-scale roughness produces velocity fluctuations and vortex shedding that were insensitive to small freestream perturbations at this Reynolds number. The authors speculate that vortex shedding from this type of roughness may be an absolute instability, which would be self-sustaining. So the shedding may generate the disturbance field that triggers it's own instability.

3.2. Roughness Reynolds number comparison

The experimental study by Bragg et al. [20] also investigated the area of critical roughness Reynolds number, ($Re_{k,\mathrm{crit}}$), for an airfoil leading edge with surface roughness. In this work they

showed that values of $Re_{k,\mathrm{crit}}$ are larger than the classic critical value of 600 when the surface roughness protrudes through the boundary layer. The reader is referred to this work for a more detailed discussion on the testing of the critical roughness Reynolds number and the related flow parameters.

Bragg et al. [20] took measurements behind several isolated three-dimensional hemispherical roughness elements for the determination of the $Re_{k,\mathrm{crit}}$ values on the leading edge of the NACA 0012. The current study focuses on the isolated roughness element with a height of 0.5 mm. As with the distributed roughness, the isolated roughness used in this study matches the basic geometry of the experiment, but is only two-dimensional.

In the experiment, the isolated roughness elements were placed at various locations downstream of the leading edge ranging from an arc length of 8–25 mm. At a fixed location and roughness height, the chord Reynolds number was increased to determine the critical roughness Reynolds number for that particular roughness geometry. The definition of the roughness Reynolds number is given by

$$Re_k = \frac{U_k k}{v}, \tag{23}$$

where k is the roughness height and U_k is the velocity at the top of the roughness element in an undisturbed boundary layer. So the main factors controlling the value of the Re_k is roughness height, location of roughness and the freestream velocity. In the present study, two leading-edge roughness locations were used with constant hump height ($h = 0.5$ mm) that fall within the test range used by Bragg et al. [20]. A schematic of the isolated roughness is shown in Fig. 22. At a given location, the freestream Reynolds number was increased to increase the value of Re_k.

Fig. 23 summarizes the results of the roughness Reynolds number study. Included in this plot are the experimental results from Bragg et al. [20] for the critical roughness Reynolds number values. Using the two leading-edge locations for the isolated roughness and varying the value of

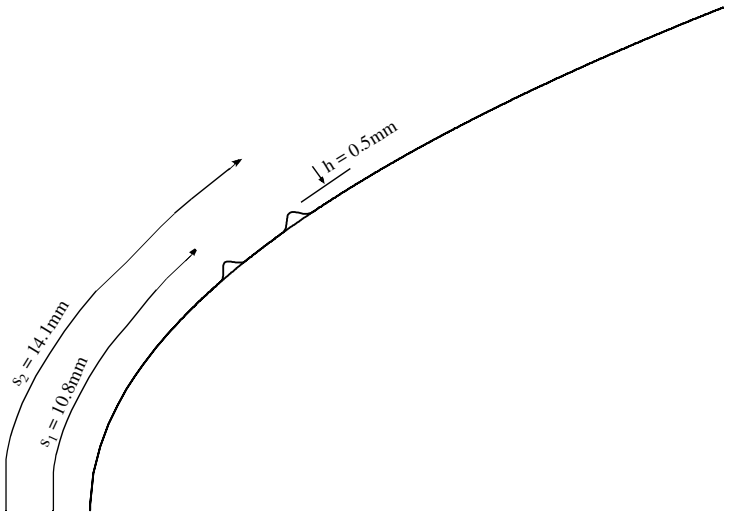


Fig. 22. Schematic diagram of the roughness locations used for the roughness Reynolds number (Re_k) calculations.

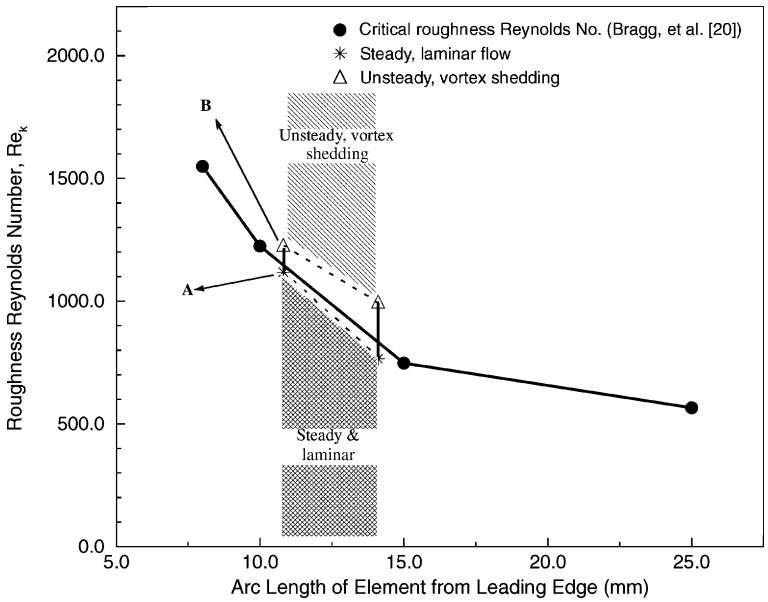


Fig. 23. Determination of Re_k for steady, laminar flow and unsteady, vortex-shedding.

Re_k, the simulation was used to determine whether the resulting flow was steady and laminar, or unsteady with vortex shedding. Therefore, at each roughness location, the plot shows the point where the computed flow is steady and laminar. Everywhere below that point the flow is also steady and laminar, as verified by further computations. Likewise, a point is shown at a higher Re_k value where the flow was unsteady with vortex shedding. Everywhere above that point the flow remains in this state. This results in a bracketed area of unsteady, vortex shedding flow and a bracketed area of steady, laminar flow. As the plot shows, these two areas straddle the $Re_{k,\mathrm{crit}}$ results of Bragg et al. [20]. This is a possible indication that there is a correlation between the observed unsteady, vortex-shedding as calculated by the current algorithm and the roughness-induced transitional flow seen in the experiments for this type of large-scale roughness.

Two of the points, labeled "A" and "B", (one steady/laminar and the other unsteady/vortex-shedding) from Fig. 23 have been selected to help visualize the flow structure at these flow conditions. Fig. 24 shows a streamfunction plot of "point A", which is the steady, laminar flow. Fig. 25 corresponds to the flow at "point B" in Fig. 23, which is the unsteady flow with vortex shedding. For clarity, the flow directly behind the roughness is displayed with streamfunction contours and the flow downstream of the roughness uses vorticity contours to visualize the flow; both snapshots in time. This clearly shows the vortex shedding downstream of the isolated roughness element.

4. Summary of findings

In this study, a full two-dimensional Navier–Stokes algorithm is used to investigate unsteady viscous flow past an airfoil leading edge with large-scale surface roughness that is characteristic of

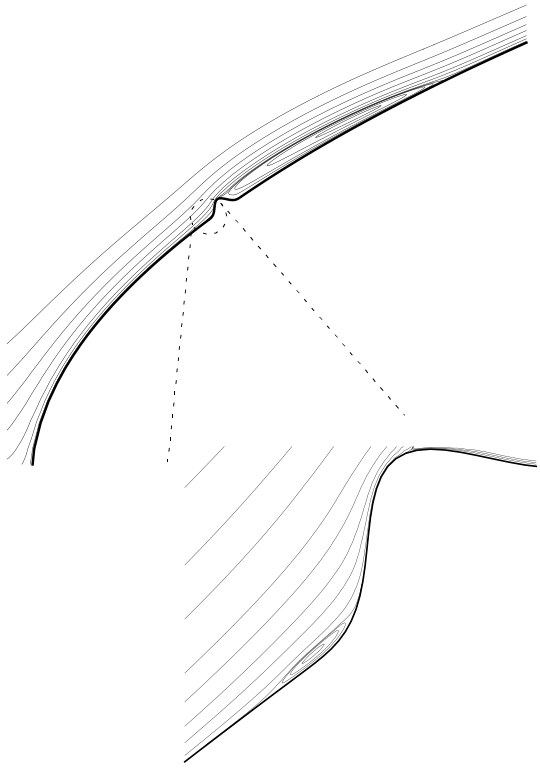


Fig. 24. Streamfunction plot for "point A" in Fig. 23, showing steady, laminar flow.

early ice accretion. The focus of this work was a detailed flow analysis of the roughness-induced velocity fluctuations and vortex shedding. The results have been compared to experimental data on roughness-induced transition for the same roughness geometries. The key findings are summarized as follows:

- The current Navier–Stokes algorithm is capable of predicting the unsteady velocity fluctuations and vortex shedding induced by the large-scale roughness.
- The calculated fluctuation intensity values show reasonable correspondence with the experimental turbulence intensity values for the roughness and Reynolds numbers investigated.
- For the roughness Reynolds number, the region between the calculated steady laminar flow and the unsteady roughness-induced vortex shedding brackets the experimental critical roughness Reynolds number values.

The authors speculate that there is a possible correlation between the observed unsteady, vortex-shedding as calculated by the current algorithm and the onset of the roughness-induced transitional flow seen in the experiments for this type of large-scale roughness. Studies are currently underway to determine the role of three-dimensionality in this process, but such considerations are beyond the scope of the current study.

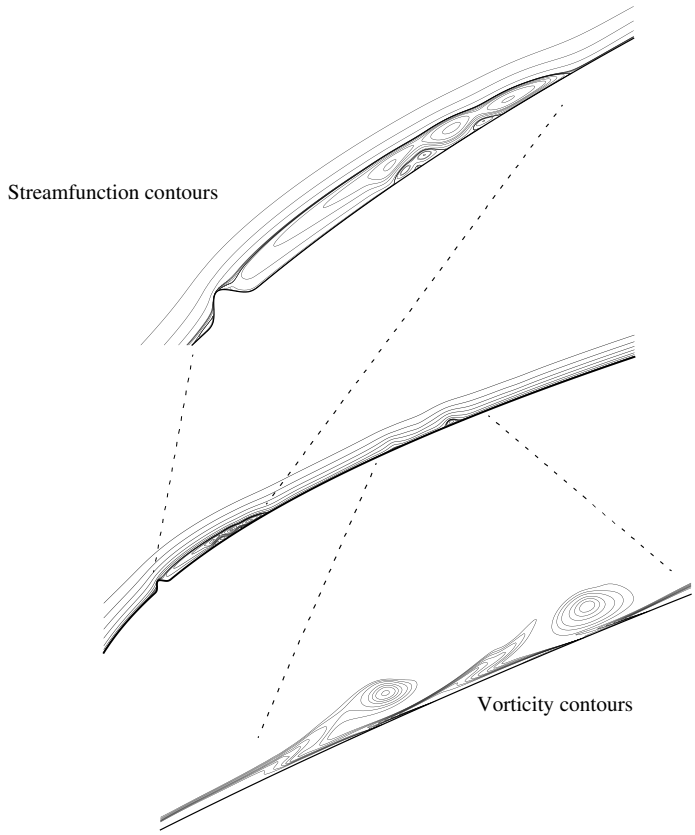


Fig. 25. Streamfunction and vorticity contours for "point B" in Fig. 23, showing unsteady, vortex shedding.

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